

FINAL REPORT: ANIME DATABASE

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Section: 3

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Final Project – Spring 2024

Introduction:

The animation industry, particularly within the realm of anime production, thrives on creativity, meticulous planning, and efficient execution. In this context, the development and management of a comprehensive database tailored specifically for anime production companies become imperative. Through three successive stages of analysis and design, we have meticulously crafted a database solution tailored to meet the multifaceted needs of such companies.

Our journey began in Stage 1, where we immersed ourselves in the intricacies of the anime production landscape. By dissecting industry requirements and company demands, we elucidated the core functionalities and target audience for our database. From project management to user accounts, ratings, and reviews, every aspect was meticulously analyzed to ensure our database would serve as a versatile and indispensable tool for anime enthusiasts and industry professionals alike.

In Stage 2, our focus shifted to the foundational elements of our database: the key entities, attributes, and relationships essential for effective operations management. From departments specializing in unique animation genres to the intricate details of anime projects, staff members, episodes, tasks, contracts, and viewer feedback, we meticulously delineated each component, laying the groundwork for a robust and cohesive database schema.

Now, in Stage 3, armed with a deep understanding of the requirements and structure, we transition into the implementation phase. Through the utilization of SQL codes, we aim to translate our conceptual framework into a functional database system capable of seamlessly orchestrating the complex interplay of projects, personnel, and production processes within an anime production company.

Our ultimate goal is to empower anime production companies with a powerful tool that not only streamlines operations but also fosters creativity, collaboration, and innovation. Whether it's anime enthusiasts exploring new series or industry professionals seeking insights and analysis, our database solution endeavors to cater to a diverse range of users, serving as a cornerstone in the vibrant world of anime production.

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Stage 1:

The case study:

The following database is designed to help anime production companies or studios for the purpose of tracking its ongoing projects, employees and the related information. This database should serve to handle the operations of an anime production company like project management, employee's data, and production details. After analyzing the industry requirements and the company's demands, the following requirements were developed for this database:

- The anime production company is divided into departments, with each focused on a unique genre or type of animation. Every department has a unique name, a unique id, a central office with contact details, and is managed by an executive producer. Each section manages a number of anime projects.
- Anime projects are the company's core business. Each project has a unique title, a unique project code, a genre (such as action, romance, fantasy, etc), and a brief description. The database also includes information about the project's primarily director, scriptwriter, and lead character designer.
- The database keeps track of all staff members, including animators, voice actors, and production helpers. Every staff member has a unique identifier, first and last name, contact information, and a specific job within the company. In addition, every staff member is assigned to a certain department.
- For animators, the database includes information such as their animation style, notable works, and the date they joined the company. Voice actors' information includes their vocal range, prominent roles, a unique identifier, and debut date. Production helpers have specific roles and responsibilities.
- The database also stores information on the episodes of each anime project. Each episode is assigned a unique number, title, brief summary, and air date. Episodes are tied with a certain project and the staff members who worked on it.
- Production schedules are critical for completing tasks by deadline. The database contains information on the beginning and end dates of each production process, including pre-production, animation, voice recording, and post-production.
- Contracts with other collaborators, such as music producers or advertising artists, are stored in the database. This comprises contract start and end dates, payment information, and which project(s) they are affiliated with.
- The database collects viewers and feedback data for each episode, which may be used to analyze performance and make future decisions.

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➤ Functionalities of this database:

❖ Basic Functions:

1. **Anime Database:** This website maintains a comprehensive database of anime series, including titles, genres, summaries, and release dates.
2. **Search and Filtering:** Users can use keywords, genres, and other criteria to find specific anime series or explore content in their preferred categories.
3. **User Accounts:** Users can establish accounts to customize their experience, monitor their watched series, and have access to additional services.
4. **Ratings and Reviews:** Users can rate and write reviews on anime series, adding to the gathering of audience ratings.

❖ Target Audience and Benefits:

The Anime Database with Advanced Animation Display serves a variety of users, including anime fans, casual watchers, and industry professionals. Here are a few examples of how various users may benefit from accessing this database:

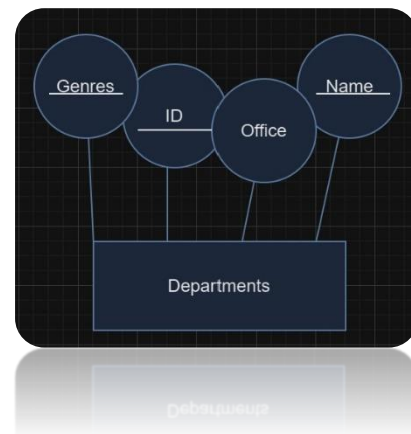
1. **Anime Enthusiasts:** Its enhanced animation display allows anime fans to discover new series and make informed selections about what to watch next. The dynamic animations provide an enhanced understanding of each series' visual appeal and animation quality, allowing fans to personalize their anime viewing experiences.
2. **Casual Viewers:** Casual viewers who are new to anime can benefit from the visual representation provided by the sophisticated animation display. It allows them to rapidly analyze the style and tone of various programs, making it easier to select content that suits their preferences and interests.
3. **Industry Professionals:** Anime producers, animators, and industry professionals can use this database to do research and analysis. The advanced animation display enables them to research various animation techniques, art styles, and trends in the anime business, thereby enhancing their creative work and decision-making processes.

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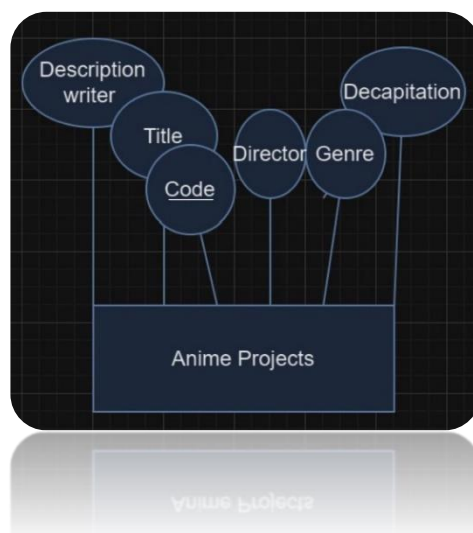
Stage 2:

The anime production company's database contains a number of essential key entities and attributes required for efficient operations management, along with relationships between these entities.

- Key Entities, Attributes, and Subclasses:
 - The first key entity is Departments, with the attributes of each specializing in a unique genre or type of animation. These departments are characterized by unique names, identifiers, central office details.

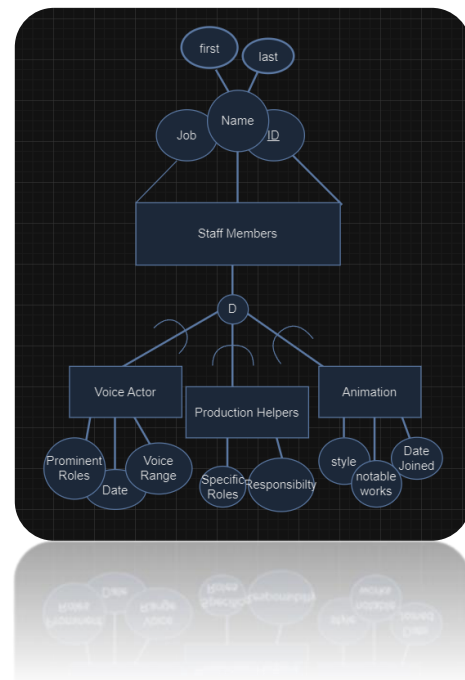


- The company's primary focus is on Anime Projects, which are identified by unique names, project codes, and genres like action, romance, or fantasy, along with brief descriptions. Each project is associated with key personnel, including directors, scriptwriters, and lead character designers, which are essential attributes of this second key entity.

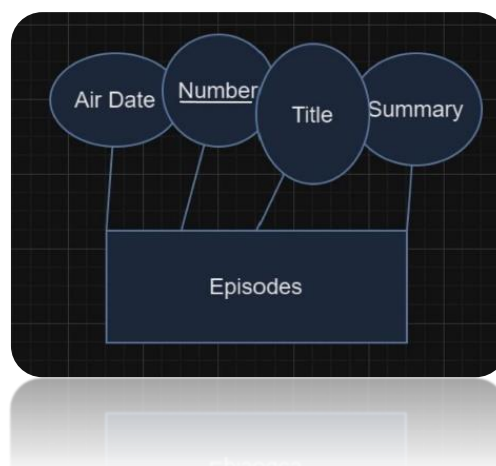


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- The third key entity is Staff Members, including the subclasses animators, voice actors, and production helpers, forming another crucial aspect of the company's structure. Each staff member is assigned a unique identifier, possesses contact information, and has a specific job within the company. Additionally, every staff member is affiliated with a certain department. For animators, attributes stored in the database include their animation style, notable works, and the date they joined the company. Voice actors' information comprises their vocal range, prominent roles, a unique identifier, and date. Production helpers are characterized by their specific roles and responsibilities, which are essential attributes of this entity.

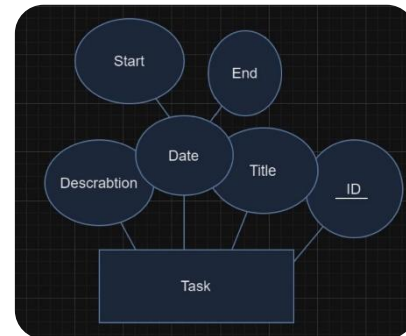


- The fourth key entity is Episodes. Each episode is assigned a unique number, title, brief summary, and air date, constituting essential attributes of this entity.

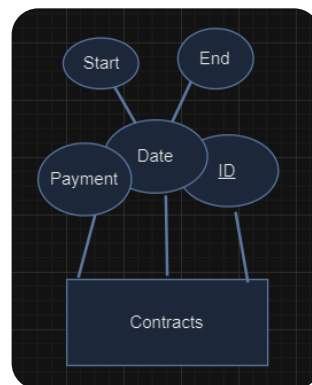


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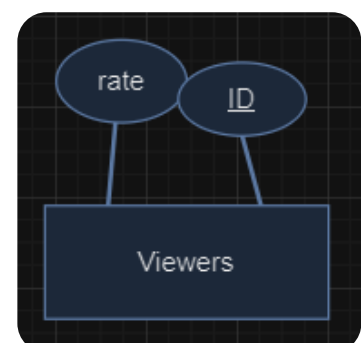
- The fifth key entity is Tasks within Episodes. Attributes of this entity include the beginning and end dates of each task, a unique identifier, title, and description of the task associated with the episode.



- The sixth key entity is Contracts with Collaborators, such as music producers or advertising artists. Attributes of this entity include contract start and end dates, as well as payment information and a unique identifier.



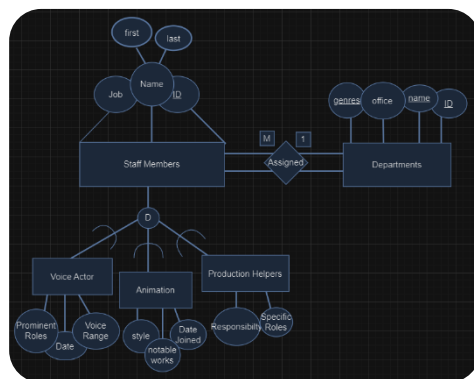
- The seventh key entity is Viewer and Feedback for Episodes. Attributes of this entity include a unique identifier and the rating of the episodes.



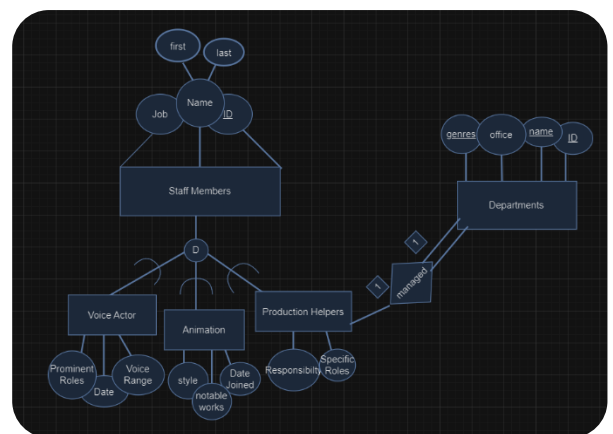
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- Relationships between entities:

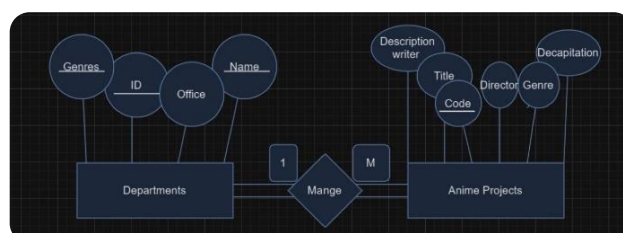
- Every staff member is assigned to a certain department.



- Each department is managed by an executive producer.

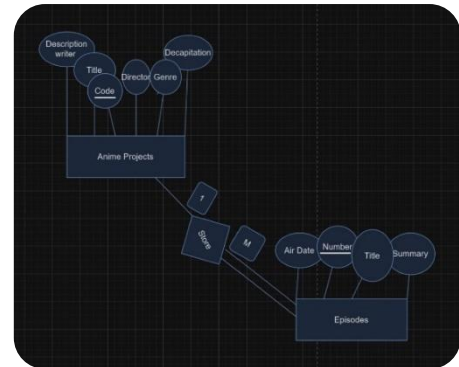


- Each department is managed to a number of anime projects.



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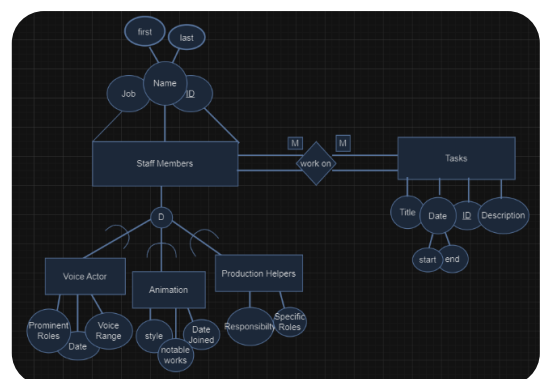
- Anime projects store episodes.



- Every episode has critical tasks.

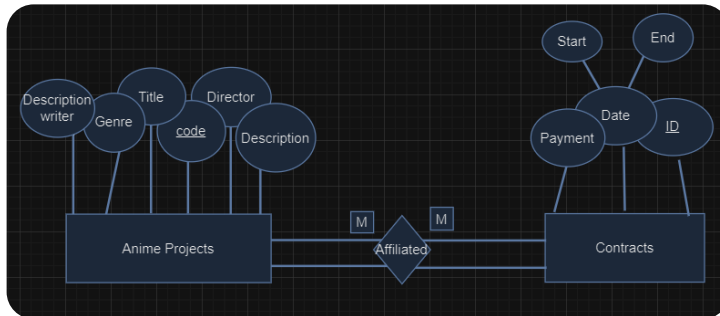


- Staff work on tasks.

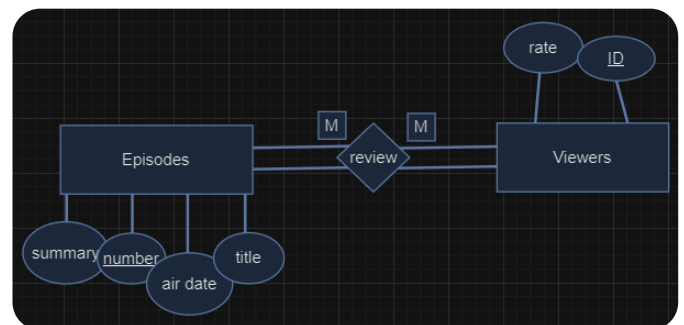


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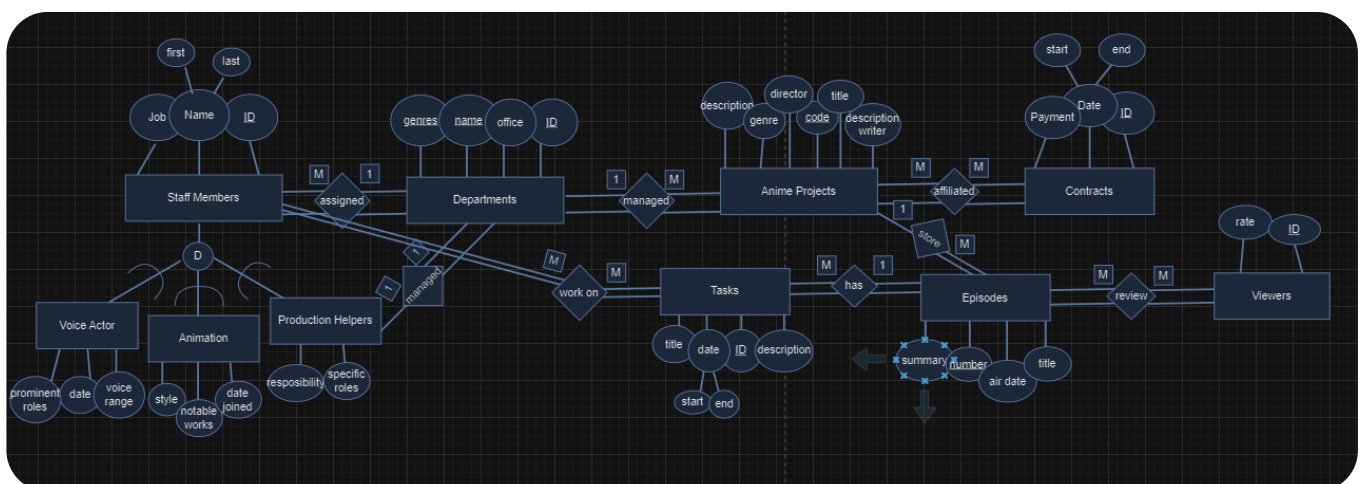
- Contracts are affiliated with which project(s).



- Each episode collects viewers and feedback.

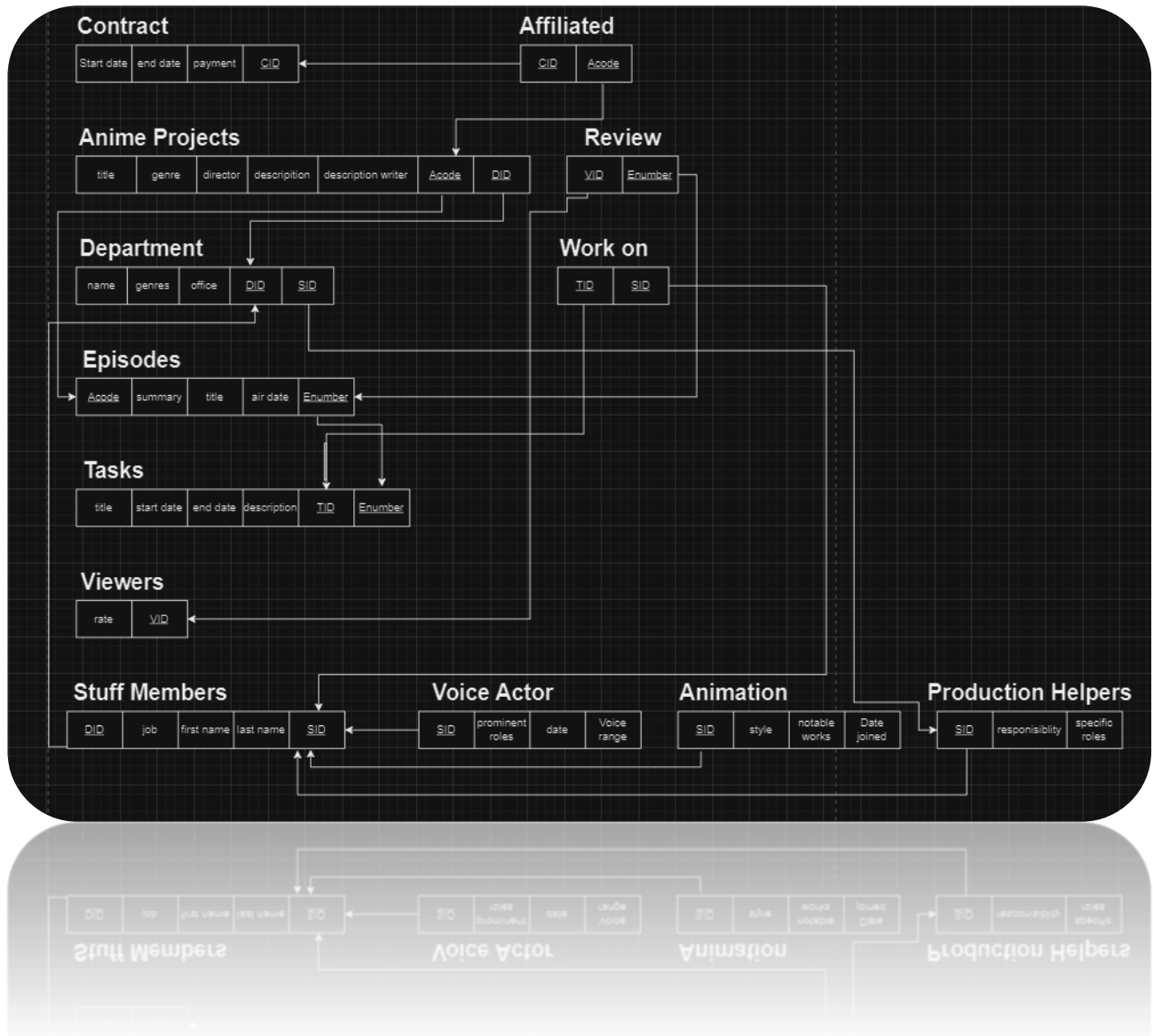


• EER Diagram:



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- Relational Schema:



Stage 3:

In this stage of our Anime Database System project, significant progress has been made in establishing the groundwork for efficient anime production management. Each table in our MySQL database has been meticulously crafted to capture crucial aspects of anime production. From animation

studios to voice actors, contracts, and individual episodes, every facet of the production process has been carefully considered. In the following sections, we will

Final Project – Spring 2024

delve into detailed descriptions of each database table, outlining the specific attributes added to facilitate comprehensive data organization and management.

◆ Affiliated

The `affiliated` table consists of two columns, `CID` and `Acode`, forming a composite primary key. It establishes two foreign key constraints: `CID` referencing the `CID` column in the `contract` table, and `Acode` referencing the `Acode` column in the `anime\_projects` table.

```
MariaDB [finalproject]> desc affiliated;
+-----+-----+-----+-----+-----+-----+
| Field | Type   | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CID   | int(11)| NO   | PRI | NULL    |       |
| Acode | int(11)| NO   | PRI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.021 sec)
```

◆ Animation

The `animation` table includes columns for `SID`, `astyle`, `Notableworks`, and `Date\_joined`. `SID` is the primary key, and a foreign key constraint ensures its integrity by referencing the `SID` column in the `stuffmembers` table.

Field	Type	Null	Key	Default	Extra
SID	int(11)	NO	PRI	NULL	
astyle	varchar(30)	YES		NULL	
Notableworks	varchar(30)	YES		NULL	
Date_joined	date	YES		NULL	

◆ Anime projects

The `anime\_projects` table holds information on anime projects, with columns for `Atitle`, `Agenres`, `Adirector`, `Adescription`, `WriterDesc`, `ADID`, and `Acode`. `Acode` serves as the primary key, while a foreign key constraint on `ADID` maintains integrity by linking to the `DID` column in the `department` table.

Field	Type	Null	Key	Default	Extra
Atitle	varchar(50)	YES		NULL	
Agenres	varchar(50)	YES		NULL	
Adirector	varchar(30)	YES		NULL	
Adescription	varchar(200)	YES		NULL	
WriterDesc	varchar(200)	YES		NULL	
ADID	int(11)	YES	MUL	NULL	
Acode	int(11)	NO	PRI	NULL	

Final Project – Spring 2024

◆ Atasks

The `atasks` table tracks anime production tasks, featuring columns for `Title`, `s\_date`, `e\_date`, `Task\_desc`, `TID`, and `Enumber`. Its primary key consists of `TID` and `Enumber`, ensuring task uniqueness within episodes. A foreign key constraint on `Enumber` maintains data integrity by linking to the `Enumber` column in the `episodes` table.

Field	Type	Null	Key	Default	Extra
Title	varchar(50)	YES		NULL	
s_date	date	YES		NULL	
e_date	date	YES		NULL	
Task_desc	varchar(200)	YES		NULL	
TID	int(11)	NO	PRI	NULL	
Enumber	int(11)	NO	PRI	NULL	

◆ Contract

The `contract` table stores anime production contracts, with columns for `Sdate`, `Edate`, `payment`, and `CID`. `CID` is the primary key.

Field	Type	Null	Key	Default	Extra
Sdate	date	YES		NULL	
Edate	date	YES		NULL	
payment	varchar(30)	YES		NULL	
CID	int(11)	NO	PRI	NULL	

◆ Department

The `department` table manages anime production departments, containing columns for `Dname`, `Dgenres`, `Doffice`, `DID`, and `SID`. `DID` serves as the primary key, uniquely identifying each department. A foreign key constraint on `SID` links it to the `SID` column in the `stuffmembers` table.

Field	Type	Null	Key	Default	Extra
Dname	varchar(30)	YES		NULL	
Dgenres	varchar(50)	YES		NULL	
Doffice	varchar(30)	YES		NULL	
DID	int(11)	NO	PRI	NULL	
SID	int(11)	YES	MUL	NULL	

Final Project – Spring 2024

◆ Episodes

The `episodes` table tracks anime episodes with columns for `Acode`, `ESummary`, `Etitle`, `AirDate`, and `Enumber`. The primary key is formed by `Enumber` and `Acode`, uniquely identifying each episode. A foreign key constraint on `Acode` maintains consistency with the `anime\_projects` table.

Field	Type	Null	Key	Default	Extra
Acode	int(11)	NO	PRI	NULL	
ESummary	varchar(200)	YES		NULL	
Etitle	varchar(50)	YES		NULL	
AirDate	date	YES		NULL	
Enumber	int(11)	NO	PRI	NULL	

◆ Production_helpers

The `production\_helpers` table manages production roles and responsibilities with columns for `SID`, `responsibility`, and `specific\_roles`. `SID` is the primary key, linking to `stuffmembers` for consistency.

Field	Type	Null	Key	Default	Extra
SID	int(11)	NO	PRI	NULL	
responsibility	varchar(30)	YES		NULL	
specific_roles	varchar(30)	YES		NULL	

◆ Review

The `review` table logs viewer reviews for episodes, with `V\_ID` and `E\_number` as primary key columns. Foreign keys tie `E\_number` to `Enumber` in `episodes` and `V\_ID` to `VID` in `viewers`.

Field	Type	Null	Key	Default	Extra
V_ID	int(11)	NO	PRI	NULL	
E_number	int(11)	NO	PRI	NULL	

Final Project – Spring 2024

◆ StaffMembers

The `stuffmembers` table manages staff details, including job roles, first names, last names, and unique identifiers (`SID`). The primary key is `SID`, and `De\_DID` links to `DID` in the `department` table for department association.

Field	Type	Null	Key	Default	Extra
S_job	varchar(30)	YES		NULL	
Fname	varchar(30)	YES		NULL	
Lname	varchar(30)	YES		NULL	
SID	int(11)	NO	PRI	NULL	
De_DID	int(11)	YES	MUL	NULL	

◆ Viewers

The `viewers` table records viewer ratings with a `rate` column and unique identifiers (`VID`). The primary key is `VID`, ensuring each viewer has a distinct identifier.

Field	Type	Null	Key	Default	Extra
rate	varchar(10)	YES		NULL	
VID	int(11)	NO	PRI	NULL	

◆ VoiceActor

The `voiceactor` table manages voice actor details, with columns for roles, start dates (`Vdate`), and availability ranges (`Vrange`). The primary key is `SID`, linked to `SID` in the `stuffmembers` table.

Field	Type	Null	Key	Default	Extra
SID	int(11)	NO	PRI	NULL	
roles	varchar(30)	YES		NULL	
Vdate	date	YES		NULL	
Vrange	varchar(30)	YES		NULL	

Final Project – Spring 2024

♦ Work_on

The `work\_on` table manages task assignments with columns for `TID` and `SID`, forming a composite primary key. Foreign keys link `TID` to `TID` in `atasks` and `SID` to `SID` in `stuffmembers` for data consistency.

Field	Type	Null	Key	Default	Extra
TID	int(11)	NO	PRI	NULL	
SID	int(11)	NO	PRI	NULL	

♦ For an example, we inserted data in (Anime_projects) table:

```
MariaDB [finalproject]> INSERT INTO anime_projects
-> VALUES ('Hotarubi no Mori e', 'Drama, Supernatural', 'Takahiro Omori', 'A six-year-old girl, Hotaru, meets Gin, a boy cursed to disappear if touched, in a forest. \ Despite the curse, they form a close friendship.', 'Yuki Midorikawa', 2, 1);
```

```
***** 1. row *****
atitle: Hotarubi no Mori e
agenres: Drama, Supernatural
adirector: Takahiro Omori
adescription: A six-year-old girl, Hotaru, meets Gin a boy cursed to disappear if touched
in a forest. Despite the curse
they form a close friendship.
writerdesc: Yuki Midorikawa
adid: 2
acode: 1
***** 2. row *****
scode: 1
sqid: 1
```

In summary, inserting data into MySQL tables is a straightforward process, enabling us to effortlessly add information and keep our database updated. This method can be applied uniformly across all tables, ensuring efficient data management for various applications.

Final Project – Spring 2024

Implementation details:

For our implementation, we opted to utilize Python as the primary language for developing the graphical user interface (GUI) to interact with the user. Leveraging the capabilities of Python, we crafted a user-friendly interface to streamline the process of inputting information into our anime database system. Visual Studio Code served as our integrated development environment (IDE), providing a robust platform for coding and testing our application. The GUI seamlessly connects to a MySQL database, facilitating the storage and retrieval of anime-related data. Users can effortlessly input information through the GUI, and the data is promptly transferred to the MySQL backend, ensuring efficient management of the anime database system. This integration of Python, Visual Studio Code, and MySQL empowers users to interact with the database system in a straightforward and intuitive manner, enhancing their overall experience.

Additionally, to ensure easy access and collaboration among team members, we've upload both the SQL database and the Python GUI code onto Dropbox. The SQL database file contains all the essential data required for our anime database system :

<https://www.dropbox.com/scl/fi/vookkqoygq2mcmfnd5so3/finalproject.sql?rlkey=04hfjo5x5fjfm4yl7gabvkm5&st=0wkdrh32&dl=0>

Likewise, the Python code for the GUI implementation :

<https://www.dropbox.com/scl/fi/9wyutjeh4gr85gp52aex4/animeuchiha.sln?rlkey=07b4y13aqcg7u8n24a3qiu58a&st=v51wo9kv&dl=0>

By hosting these resources on Dropbox, we aim to streamline access and foster seamless collaboration among team members.

Final Project – Spring 2024

Conclusion:

In conclusion, our journey through the development of the Anime Database has been a testament to the power of collaboration, perseverance, and continuous learning. From the initial stages of analysis and design to the implementation phase, each step was marked by diligent effort, innovative problem-solving, and an unwavering commitment to excellence. Our experiences have not only enhanced our technical skills but also fostered a sense of camaraderie and mutual support within our team. The challenges we faced, particularly in integrating Python with SQL, served as opportunities for growth and strengthened our resolve to overcome obstacles together. As we look to the future, we are filled with optimism and enthusiasm for the possibilities that lie ahead. Armed with the knowledge and insights gained from this project, we are well-equipped to tackle even greater challenges and push the boundaries of database development. Ultimately, our Anime Database project represents more than just a culmination of technical achievements. It embodies our shared passion for innovation, our dedication to continuous improvement, and our belief in the transformative power of teamwork. With each step forward, we reaffirm our commitment to excellence and our vision of creating impactful solutions that empower industries and enrich lives.

References:

1. Brain's Base. (2011). *Hotarubi no Mori e* [Film].
https://myanimelist.net/anime/10408/Hotarubi_no_Mori_e
2. MyAnimeList. (2013). *Shingeki no Kyojin* [MyAnimeList page]. MyAnimeList.
https://myanimelist.net/anime/16498/Shingeki_no_Kyojin
3. IMDb. (2011). *To the Forest of Firefly Lights* [IMDb page]. IMDb. Retrieved from
<https://www.imdb.com/title/tt2061702/>